THE PLANT ARCHITECTURE INVERSION EXPERIMENT

Coherence Gradient Field Specification Produces Pre-Specified Architectural Inversion in *Raphanus sativus*

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The organism is not the output of a genome alone. It is the output of a genome operating within a coherence gradient field. Change the field — the genome executes a different program. Same genome. Different field. Different organism. Placing a radish seed in an inverted coherence gradient field — mineral water above, UV-A light below — produces root growing upward, shoot growing downward. Conditions derived geometrically from the attractor. Prediction locked before the experiment. $29–54 total cost. Primary result in 72 hours. Pre-registered 2026-03-13.

The Theorem

Standard model: Genome → Organism.
Complete model: Genome + Coherence Field → Organism.

Standard plant architecture (root down, shoot up) is not genetically fixed. It is the attractor the genome resolves to in the standard gradient field. **Invert the field before germination: the genome resolves to the inverted attractor.** No genetic modification. No mutation. One generation.

Root Navigates toward water, minerals, darkness.
Shoot Navigates toward light, CO_2.

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Pre-Registered Prediction

In the **EXP** chamber (inverted gradient field):

Root Grows **upward.** Hydrotropism overrides gravitropism.
Shoot Grows **downward.** Phototropism inverted by field geometry.
AIS = 2 confirmed at T=72 h, maintained at T=168 h.

AIS = 2: root >135°, shoot <45° from vertical, measured by protractor against transparent chamber wall.

C1 control: AIS = 0 (standard architecture).

C4 control: root upward, reproduces PNAS 2025.

What Confirmation Establishes

1. Field authorship of architecture. Same genome, inverted field, inverted organism. Architecture is a field output, not a fixed genetic output. First demonstration from a pre-specified, locked prediction.

2. Attractor geometry is real and navigable. The plant navigated to the attractor specified by the field. Field is the instruction set. Genome is the build capacity.

3. No genome editing required. Specify the field. The genome executes. No CRISPR. No off-target effects. One generation.

The Four Chambers

ID	Setup	Expected
EXP	Water above. UV-A below. Black cover above.	AIS = 2. Root up. Shoot down.
C1	Moist agar. Ambient light above.	AIS = 0. Standard. Baseline.
C3	UV-A below. No water. Black cover.	Root down. Shoot down.
C4	Water above. Ambient light above.	Root up. Shoot up. Reproduces PNAS 2025.

C3: phototropic component alone. C4: hydrotropic component alone. EXP: both combined. AIS = 2 requires *both* components simultaneously.

Materials and Timeline

Item	Cost	Source
Radish seeds	$2–4	Garden store
Agar powder	$3–8	Grocery store
Mineral salts or mineral water	$3–8	Grocery/hydro
Transparent chambers (5×)	$3–8	Hardware
UV-A LED strip ∼365 nm	$8–15	Amazon
Distilled water	$1–2	Grocery store
Black cardboard	$1–3	Dollar store
Total	**$21–48**	

No BSL rating. No credentials. No specialist equipment.

2–3 days preparation. Primary result T = 72 h. Full confirmation T = 168 h (Day 7).

Chambers transparent — observable by eye throughout.

The field came first. The genome was always listening. The architecture was never fixed. The inversion was always possible.

github.com/Eric-Robert-Lawson/attractor-oncology · doi.org/10.5281/zenodo.19040399 · ORCID: 0009-0002-0414-6544